INCH-POUND

A–A–59590/24A
17 October 2007
SUPERSEDING
A–A–59590/24
6 September 2002

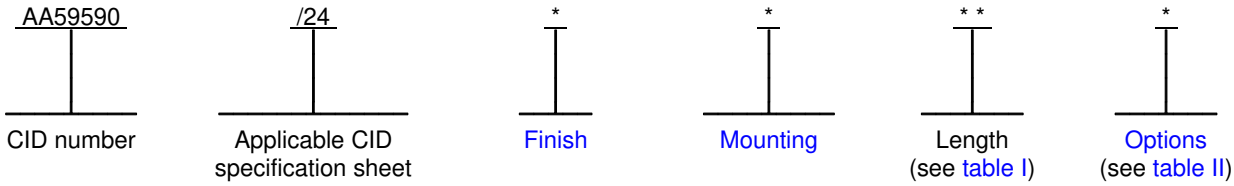
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE .500 X .450 INCH BODY SIZE, WITH VISUAL LOCK INDICATION AND SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes).



Example: AA59590/24ET48E is the PIN for a nickel plated, 4.8 inches (122 mm) long card holder with a visual lock indication and a screw self-locking element. The card holder also features two tapped 0-80 UNF mounting holes for use with fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position and provide high resistance to loosening due to shock and vibration. Card holders shall also provide maximum thermal transfer from the circuit card assembly to the cold plate or heat sink surfaces. The screw retention feature is intended to prevent unintentional disassembly of the card holder.

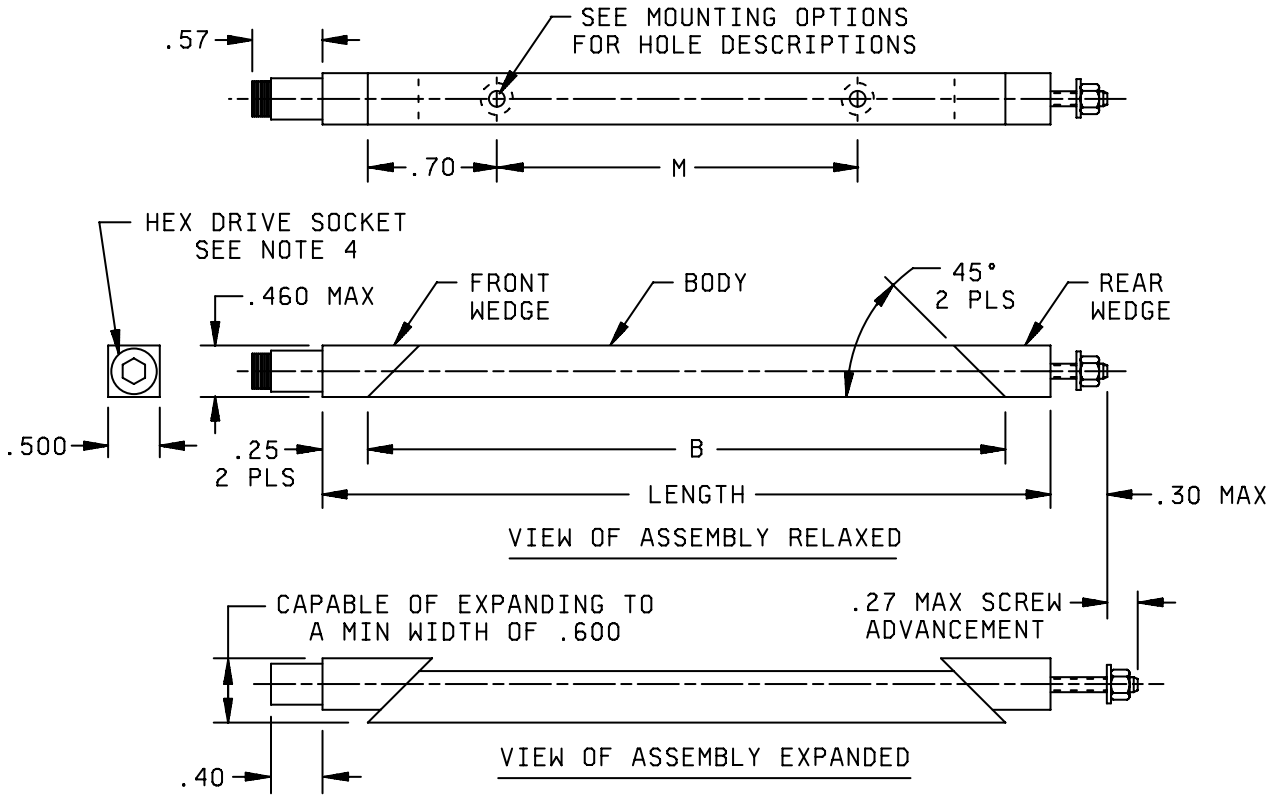
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein (see figures 1, 2, 3, 4, and table I) and meet the general requirements of CID A–A–59590.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .525 inch (13.3 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 31 inch-pounds (3.5 N-m) for assemblies using no options or option “W” and 34 to 36 inch-pounds (3.8 to 4.1 N-m) for assemblies using options “D” or “E”.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.156	3.96	.27	6.9	.40	10.2	.500	12.7	.600	15.24
.25	6.4	.30	7.6	.460	11.68	.57	14.5	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The tolerance for the mounting hole spacing is ±.005 inches (0.13 mm).
4. The across flats dimension for hex drive socket shall be .156 inch (3.96 mm) for mounting options “J”, “N”, “T”, and “V”. The across flats dimension for hex drive socket shall be a metric dimension of 3.0 mm (.118 inch) across flats for mounting option “G”.

FIGURE 1. Relaxed and expanded dimensions, basic configuration.

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Screw retention (locknut) (see figure 2). The screw retention feature is intended to prevent unintentional disassembly of the card holder shall be as specified in figure 2. The locknut shall be fabricated of a corrosion resistant material.

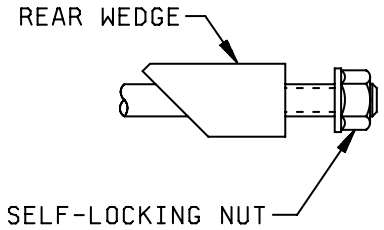


FIGURE 2. Screw retaining self-locking nut details.

Visual lock indicator (see figure 3). Card holders shall have a visual indicator to show when the card holder assembly is in its relaxed (unlocked) state. When the card holder assembly is in the relaxed state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened so that the assembly is in the expanded (locked) position, this red band shall be concealed.

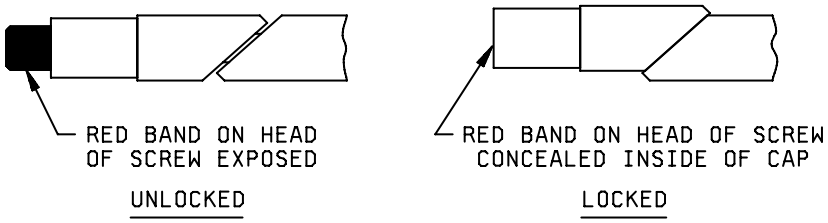


FIGURE 3. Visual lock indicator details.

Finish. The finish designator shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: “B” (black anodize), “C” (chemical film), “E” (electroless nickel), or “H” (hard black anodize).

Mounting. The mounting designators shall be as specified in A–A–59590. Except when using mounting option “N”, two mounting holes are required on the card holders. The mounting options available for this CID specification sheet are as follows: “J” (rivet mounting holes), “N” (no mounting holes), “T” (tapped 0-80 UNF 2B holes), “V” (tapped 2-56 UNC 2B holes), or “G” (tapped M2.5 x.45 metric holes). See figure 1 and table I for mounting hole spacing requirements. Card holders using mounting option “J” may be shipped unassembled.

Counterbore holes (for rivet mounting). The holes used for rivet mounting shall be .136 inch (3.45 mm) diameter, countersunk 100 degrees by .190/.200 inch (4.83/5.08 mm) diameter through holes with an access/clearance counterbore of .190/.200 inch (4.83/5.08 mm) diameter by .390 inch (9.91 mm) deep.

Length, expanded, and relaxed dimensions. The length designator shall be as specified in A–A–59590 and the lengths available for this CID specification sheet are listed in table I. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

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TABLE I. Assembly dimensions (see figure 1).

PIN length designator	Dimension “LENGTH” ±.02 (0.5 mm)	Dimension “B” ±.03 (0.8 mm)	Dimension “M” ±.005 (0.13 mm)
28	2.8 (71 mm)	2.30 (58.4 mm)	.90 (22.9 mm)
38	3.8 (97 mm)	3.30 (83.8 mm)	1.90 (48.3 mm)
48	4.8 (122 mm)	4.30 (109.2 mm)	2.90 (73.7 mm)
58	5.8 (147 mm)	5.30 (134.6 mm)	3.90 (99.1 mm)

Options. Card holders can have the following options described in table II. See table II for the correct PIN configuration identifier. Card holders not requiring these options shall leave the options position in the PIN blank.

TABLE II. Options.

Configuration	Applicable figures	Hardware added to basic configuration
	1, 2 and 3	No changes from basic configuration
W	1, 2, 3 and 4	Lockwasher and flat washer
E	1, 2, 3 and 5	Screw self-locking element
D	1, 2, 3, 4 and 5	Lockwasher, flat washer, and screw self-locking element

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

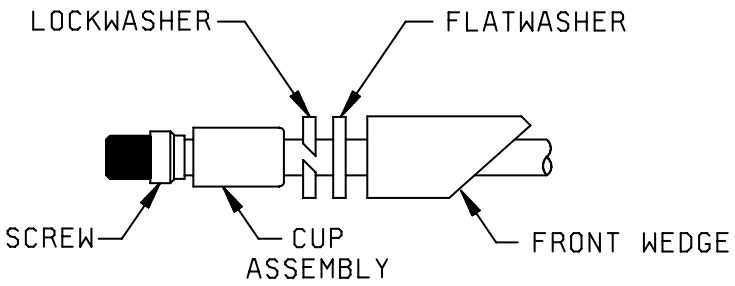


FIGURE 4. Lockwasher and flat washer details.

Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

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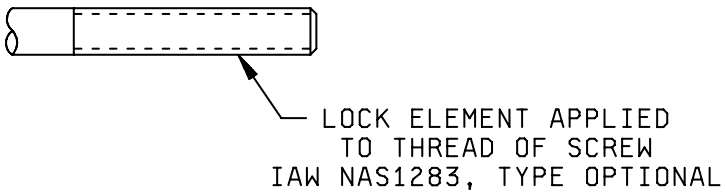


FIGURE 5. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description.

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Other Publications.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA/NAS 1283 – Fasteners, Male Threaded, Self–locking.

(Application for copies should be addressed to the Aerospace Industries Association, 1250 Eye Street, NW, Suite 1200, Washington, DC 20005–3924 or at URL: http://www.aia-aerospace.org.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

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Part number supersession data. These CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. Commercial part number supersession data.

PIN designator AA59590/24	Vendor similar designator or type part number 1/ 2/ CAGE 52094	PIN designator AA59590/24	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BJ28	VA255CR2.80HR	BJ28W	WVA255CR2.80HR
BN28	VA255CR2.80R	BN28W	WVA255CR2.80R
BT28	VA255CR2.80T0R	BT28W	WVA255CR2.80T0R
BV28	VA255CR2.80T2R	BV28W	WVA255CR2.80T2R
BG28	MVA255CR2.80TM2.5R	BG28W	MWVA255CR2.80TM2.5R
BJ28E	VA255CR2.80HLR	BJ28D	WVA255CR2.80HLR
BN28E	VA255CR2.80LR	BN28D	WVA255CR2.80LR
BT28E	VA255CR2.80T0LR	BT28D	WVA255CR2.80T0LR
BV28E	VA255CR2.80T2LR	BV28D	WVA255CR2.80T2LR
BG28E	MVA255CR2.80TM2.5LR	BG28D	MWVA255CR2.80TM2.5LR
CJ28	V255CR2.80HR	CJ28W	WV255CR2.80HR
CN28	V255CR2.80R	CN28W	WV255CR2.80R
CT28	V255CR2.80T0R	CT28W	WV255CR2.80T0R
CV28	V255CR2.80T2R	CV28W	WV255CR2.80T2R
CG28	MV255CR2.80TM2.5R	CG28W	MWV255CR2.80TM2.5R
CJ28E	V255CR2.80HLR	CJ28D	WV255CR2.80HLR
CN28E	V255CR2.80LR	CN28D	WV255CR2.80LR
CT28E	V255CR2.80T0LR	CT28D	WV255CR2.80T0LR
CV28E	V255CR2.80T2LR	CV28D	WV255CR2.80T2LR
CG28E	MV255CR2.80TM2.5LR	CG28D	MWV255CR2.80TM2.5LR
EJ28	VEN255CR2.80HR	EJ28W	WVEN255CR2.80HR
EN28	VEN255CR2.80R	EN28W	WVEN255CR2.80R
ET28	VEN255CR2.80T0R	ET28W	WVEN255CR2.80T0R
EV28	VEN255CR2.80T2R	EV28W	WVEN255CR2.80T2R
EG28	MVEN255CR2.80TM2.5R	EG28W	MWVEN255CR2.80TM2.5R

See footnotes at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/24	Vendor similar designator or type part number 1/ 2/ CAGE 52094	PIN designator AA59590/24	Vendor similar designator or type part number 1/ 2/ CAGE 52094
EJ28E	VEN255CR2.80HLR	EJ28D	WVEN255CR2.80HLR
EN28E	VEN255CR2.80LR	EN28D	WVEN255CR2.80LR
ET28E	VEN255CR2.80T0LR	ET28D	WVEN255CR2.80T0LR
EV28E	VEN255CR2.80T2LR	EV28D	WVEN255CR2.80T2LR
EG28E	MVEN255CR2.80TM2.5LR	EG28D	MWVEN255CR2.80TM2.5LR
HJ28	VHA255CR2.80HR	HJ28W	WVHA255CR2.80HR
HN28	VHA255CR2.80R	HN28W	WVHA255CR2.80R
HT28	VHA255CR2.80T0R	HT28W	WVHA255CR2.80T0R
HV28	VHA255CR2.80T2R	HV28W	WVHA255CR2.80T2R
HG28	MVHA255CR2.80TM2.5R	HG28W	MWVHA255CR2.80TM2.5R
HJ28E	VHA255CR2.80HLR	HJ28D	WVHA255CR2.80HLR
HN28E	VHA255CR2.80LR	HN28D	WVHA255CR2.80LR
HT28E	VHA255CR2.80T0LR	HT28D	WVHA255CR2.80T0LR
HV28E	VHA255CR2.80T2LR	HV28D	WVHA255CR2.80T2LR
HG28E	MVHA255CR2.80TM2.5LR	HG28D	MWVHA255CR2.80TM2.5LR

1/ The CID PINs listed are only for length designator “28”.
2/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/24	Vendor similar designator or type part number 1/ 2/ CAGE 52094
ET28E	VEN255CR2.80T0LR
ET38E	VEN255CR3.80T0LR
ET48E	VEN255CR4.80T0LR
ET58E	VEN255CR5.80T0LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.

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DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

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